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Homework 9

1.

Exercise 4.9 Show that

$$\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{-(x-\mu)^2 y/2\sigma^2\} \, \mathrm{d}x = \exp\left\{-\frac{\mu^2}{1+2\sigma^2/y}\right\}$$

by completing the square in the exponent to evaluate the integral.

We start with

$$\begin{aligned}
&\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{\frac{-(x-\mu)^2 y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{(x-\mu)^2 y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{(x^2+\mu^2-2x\mu)y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{x^2 y}{2\sigma^2} - \frac{y\mu^2}{2\sigma^2} + \frac{xy\mu}{\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})x^2 + \frac{xy\mu}{\sigma^2} - \frac{y\mu^2}{2\sigma^2}\} dx
\end{aligned}$$

With this expression equating to the form of $-ax^2 + bx + c$, we can try to complete the square as instructed, which would yield $-a(x - \frac{b}{2a})^2 + \frac{b^2}{4a} + c$. Let's try to match the format with $a = 1 + \frac{y}{2\sigma^2}$, $b = \frac{y\mu}{\sigma^2}$, and $c = -\frac{y\mu^2}{2\sigma^2}$:

$$\begin{aligned}
\frac{b}{2a} &= \frac{\frac{y\mu}{\sigma^2}}{2 + \frac{y}{\sigma^2}} \\
&= \frac{\frac{y\mu}{\sigma^2}}{\frac{2\sigma^2+y}{\sigma^2}} \\
&= \frac{y\mu}{2\sigma^2 + y}
\end{aligned}$$

and

$$\frac{b^2}{4a} = \frac{\frac{y^2\mu^2}{\sigma^4}}{4 + \frac{2y}{\sigma^2}}$$

$$= \frac{\frac{y^2\mu^2}{\sigma^4}}{\frac{4\sigma^2+2y}{\sigma^2}}$$

$$= \frac{y^2\mu^2}{4\sigma^4+2y\sigma^2}$$

So plugging in:

$$\int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2}{4\sigma^4+2y\sigma^2} - \frac{y\mu^2}{2\sigma^2}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2 - y\mu^2(2\sigma^2+y)}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2 - 2y\mu^2\sigma^2 - y^2\mu^2}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{-2y\mu^2\sigma^2}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2 + \frac{-y\mu^2}{2\sigma^2+y}\}dx$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x-\frac{y\mu}{2\sigma^2+y})^2\}dx$$

The integral is the Gaussian integral, with the form $\int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}$. Using this:

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{\pi}{1+\frac{y}{2\sigma^2}}}$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{\pi}{\frac{2\sigma^2+y}{2\sigma^2}}}$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

With the additional term in the corrected version of the problem:

$$= \frac{1}{\sqrt{\frac{2\pi\sigma^2}{y}}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

$$= \sqrt{\frac{y}{2\pi\sigma^2}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

$$
\begin{aligned}
&= \sqrt{\frac{y}{2\pi\sigma^2} \cdot \frac{2\pi\sigma^2}{2\sigma^2 + y}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \sqrt{\frac{y}{2\sigma^2 + y}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \sqrt{\frac{y}{2\sigma^2 + y}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2+y}{y}}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{\frac{-y\mu^2}{2\sigma^2 + y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{-\frac{\mu^2}{\frac{2\sigma^2+y}{y}}\} \\
&= \boxed{\frac{1}{\sqrt{\frac{2\sigma^2}{y} + 1}} \exp\{-\frac{\mu^2}{1 + \frac{2\sigma^2}{y}}\}}
\end{aligned}
$$

exactly as we sought to show in the corrected version of the problem.

2.

Exercise 4.10 Consider an Accept–Reject method for the target density f based on the instrumental distribution g.

a. Show that the Accept–Reject sample $(X_1, \ldots, X_n)$ can be associated with two iid samples, $(U_1, \ldots, U_N)$ and $(Y_1, \ldots, Y_N)$, with corresponding distributions $\mathcal{U}_{[0,1]}$ and g; N is the stopping time associated with the acceptance of n variables Y_j.

b. Deduce that the corresponding estimator of $\mathbb{E}_f[h(X)]$ based on $(X_1, \ldots, X_n)$ can be written as

$$\delta_1 = \frac{1}{n} \sum_{i=1}^{n} h(X_i) = \frac{1}{n} \sum_{j=1}^{N} h(Y_j)\, \mathbb{I}_{U_j \leq w_j}\,,$$

with $w_j = f(Y_j)/Mg(Y_j)$.

We start with considering the accept-reject method for the target desntiy f based on the instrumental distribution g.

(a)

We aim to show that the Accept-Reject sample $(X_1, \ldots, X_n)$ can be associated with two iid samples, $(U_1, \ldots, U_N)$ and $(Y_1, \ldots, Y_N)$ with corresponding distributions $\text{Unif}_{[0,1]}$ and g, where N is the stopping time associated with the acceptance of n variables Y_j.

For generating samples for a target f using g, given we determine some value M such that $M \cdot g(x) \geq f(x)$ for all x, we know that the accept-reject method consists of the following steps:

- Generate $Y \sim g$
- Generate $U \sim \text{Unif}_{[0,1]}$
- If $M \cdot U \leq \frac{f(Y)}{g(Y)}$, accept Y as a draw from $f(x)$.

So from looking at this, our final sample $(X_1, \ldots, X_n)$, by definition, we are repeating this process of generating the initial Y_i and U_i N times, which is when we stop generating. Additionally, we see that by definition, we are looking to stop only when we accept n values of X_i, so we see that by definition, we can see that the sequence of n accepted values of X can indeed by associated with those independently drawn series of N U and Y values, as we sought to show.

(b)

We aim to deduce that the corresponding estimator of $E_f[h(X)]$ based on $(X_1, \ldots, X_n)$ can be written as

$$\delta_1 = \frac{1}{n}\sum_{i=1}^{n} h(X_i) = \frac{1}{n}\sum_{j=1}^{N} h(Y_j) I_{U_j \leq w_j}$$

with $w_j = \frac{f(Y_j)}{Mg(Y_j)}$.

We start with the standard estimator $\frac{1}{n}\sum_{i=1}^{n} h(X_i)$. We know that the X values in this expectation are the same as those Y values that we decided to accept in the accept-reject algorithm steps. The indicator for $U_j \leq w_j$, with the definition of w_j, is literally the same as the criteria that is used to determine whether we accept the Y value as part of the sample or not; i.e. we are taking the value and multiplying by 1 as part of our estimator if it fulfills the condition or multiplying by 0 and discarding it as part of our estimator, but still taking the sum of all values up to all N values, which was the number of values that we generated. So we took the equivalent values, but just have excess terms in the summation since we're incorporating all the rejected values but zeroing them out. Finally, we divide by n to make this a true estimator and take the average of all the accepted values as we did in the first form. These two forms are therefore equivalent.

3.

Exercise 4.15 For each of the following cases, generate random variables X_i and Y_i and compare the empirical average and Rao–Blackwellized estimator of $\mathbb{E}_f(X)$ and $\text{var}_f(X)$:

a. $X|y \sim \mathcal{P}(y)$, $Y \sim \mathcal{G}a(a,b)$ (X is negative binomial);
b. $X|y \sim \mathcal{N}(0,y)$, $Y \sim \mathcal{G}a(a,b)$ (X is a generalized t);
c. $X|y \sim \mathcal{B}in(y)$, $Y \sim \mathcal{B}e(a,b)$ (X is beta-binomial).

We aim to generate random variables X_i and Y_i and compare the empirical average and Rao-Blackwellized estimator of $E_f(X)$ and $\text{Var}_f(X)$:

(b)

$X|y \sim N(0,y), Y \sim \text{Gamma}(a,b)$, i.e. X is a generalized t:

```
> nmbSimulations = 1000
> n = 1000
> a = 5
> b = 1
>
> bruteForce_3b = function(nmbSimulations, n, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Gamma(a,b)
+     y = rgamma(n, shape = a, scale = b)
+
+     # For each Y value, simulate X|Y ~ N(0, y)
```

```
+     x = rnorm(n, mean = 0, sd = sqrt(y))
+
+     means[i] = mean(x)
+     means_squared[i] = mean(x^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Empirical results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", a =", a,
+               ", b =", b
+               ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> rao_blackwell_3b = function(nmbSimulations, n, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Gamma(a,b)
+     y = rgamma(n, shape = a, scale = b)
+
+     # Calculate conditional expectation E(X|Y)
+     # Since X|Y ~ N(0, y), we have E(X|Y) = 0
```

```
+     condExpectation = rep(0, n)
+
+     means[i] = mean(condExpectation)
+     means_squared[i] = mean(condExpectation^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Rao-Blackwellized results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", a =", a,
+               ", b =", b
+   ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> bruteForce_3b(nmbSimulations, n, a, b)
[1] "Empirical results with nmbSimulations = 1000 , n = 1000 , a = 5 ,
b = 1"
[1] "mean = 0.00101562111045003 and variance = 5.01758577452203"
> rao_blackwell_3b(nmbSimulations, n, a, b)
[1] "Rao-Blackwellized results with nmbSimulations = 1000 , n = 1000 ,
a = 5 , b = 1"
[1] "mean = 0 and variance = 0"
```

We see that the variance is set to 0 in the Rao-Blackwellized estimator. The variability in the variable generation was completely removed by using the conditional expectation instead of actually simulating X values from the normal distribution, which introduces actual variance.

(c)

$X|y \sim Bin(y), Y \sim \text{Beta}(a, b)$, i.e. X is a beta-binomial t:

```
asdf> nmbSimulations = 1000
> n = 1000
> m = 10
> a = 2
> b = 2
>
> bruteForce_3c = function(nmbSimulations, n, m, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Beta(a,b)
+     y = rbeta(n, shape1 = a, shape2 = b);
+
+     # For each Y value, simulate X|Y ~ Binomial(m, y)
+     x = rbinom(n, size = m, prob = y);
+
+     means[i] = mean(x)
+     means_squared[i] = mean(x^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
+   print(paste("Empirical results with",
```

```
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", m =", m,
+               ", a =", a,
+               ", b =", b
+   ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> rao_blackwell_3c = function(nmbSimulations, n, m, a, b) {
+
+   # Need to keep mean and mean^2 for calculating variance
+   means = rep(0, nmbSimulations)
+   means_squared = rep(0, nmbSimulations)
+
+   for (i in 1:nmbSimulations) {
+
+     # Simulate n values of Y from Beta(a,b)
+     y = rbeta(n, shape1 = a, shape2 = b);
+
+     # Calculate conditional expectation E(X|Y)
+     # Since X|Y ~ Binomial(m, y), we have E(X|Y) = m * y
+     condExpectation = m * y;
+
+     means[i] = mean(condExpectation)
+     means_squared[i] = mean(condExpectation^2)
+   }
+
+   variance = mean(means_squared) - (mean(means))^2
+
```

```
+   print(paste("Rao-Blackwellized results with",
+               "nmbSimulations =", nmbSimulations,
+               ", n =", n,
+               ", m =", m,
+               ", a =", a,
+               ", b =", b
+   ))
+   print(paste("mean =", mean(means), "and variance =", variance))
+ }
>
> bruteForce_3c(nmbSimulations, n, m, a, b)
[1] "Empirical results with nmbSimulations = 1000 , n = 1000 , m = 10 ,
a = 2 , b = 2"
[1] "mean = 5.001012 and variance = 6.998882975856"
> rao_blackwell_3c(nmbSimulations, n, m, a, b)
[1] "Rao-Blackwellized results with nmbSimulations = 1000 , n = 1000 ,
m = 10 , a = 2 , b = 2"
[1] "mean = 5.00453563874706 and variance = 4.99710120693035"
```

We see that the variance is reduced by a significant amount ($\frac{2}{7} =$ 28% or so).

4.

Exercise 4.18 A naïve way to implement the antithetic variable scheme is to use both U and $(1-U)$ in an inversion simulation. Examine empirically whether this method leads to variance reduction for the distributions (i) $f_1(x) = 1/\pi(1+x^2)$, (ii) $f_2(x) = \frac{1}{2}e^{-|x|}$, (iii) $f_3(x) = e^{-x}\mathbb{I}_{x>0}$, (iv) $f_4(x) = \frac{2}{\pi\sqrt{3}}\left(1+x^2/3\right)^{-2}$, and (v) $f_5(x) = 2x^{-3}\mathbb{I}_{x>1}$. Examine variance reductions of the mean, second moment, median, and 75th percentile.

To calculate the weights for the Rao–Blackwellized estimator of Section 4.6, it is necessary to derive properties of the distribution of the random variables in the Accept–Reject algorithm. The following problem is a rather straightforward exercise in distribution theory and is only made complicated by the stopping rule of the Accept–Reject algorithm.

(i)

(ii)

(iv)

5.

Assuming that we use antithetic variables to estimate some parameter of the standard normal distribution, we must prove that the covariance between X_i and Y_i is -1, i.e. $\text{Cov}(X_i, Y_i) = -1$.

We set up with $U \sim \text{Unif}(0, 1)$. Using the hint with the normal distribution being symmetric about 0, we know that $\Phi(z) = 1 - \Phi(-z)$ for the range of U, and equivalently that

$$x = \Phi(z) = 1 - \Phi(-z)$$

$$\Phi(-z) = 1 - x$$

$$-z = \Phi^{-1}(1 - x)$$

$$-\Phi^{-1}(-x) = \Phi^{-1}(1 - x)$$

So taking advantage of this, let's set $X = \Phi^{-1}(U)$ and $Y = \Phi^{-1}(1 - U)$ accordingly. We can then determine that

$$Y = \Phi^{-1}(1 - U) = -\Phi^{-1}(-U) = -X$$

So therefore

$$\text{Cov}(X_i, Y_i) = \text{Cov}(X_i, -X_i) = -\text{Cov}(X_i, X_i)$$

$$= -\text{Var}(X_i) = \boxed{-1}$$

just as we aimed to prove.